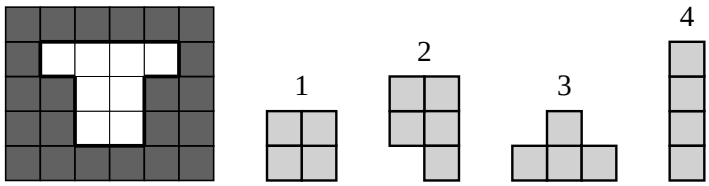
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 70989

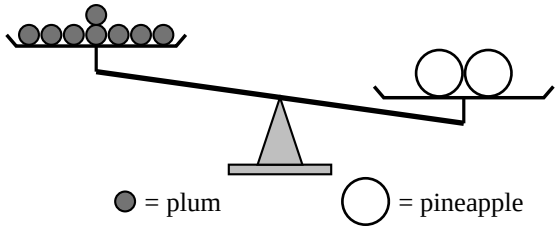
Part A: Each question is worth 3 points

1. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2 ***B.*** 1+3 ***C.*** 1+4 ***D.*** 2+3 ***E.*** 3+4

2. On one plate of a balance there are 8 plums and on the other there are 2 pineapples. When we put one more pineapple, exactly like the others, on the plate with the plums, the balance is level.



One pineapple weighs the same as how many plums?

A. 8 plums ***B.*** 7 plums ***C.*** 6 plums ***D.*** 4 plums ***E.*** 2 plums

3. Leo, Fabienne, Nora, Hana and Tom each play a different instrument: the flute, the piano, the drum, the xylophone and the harp. Nora presses keys. Leo plays an instrument with strings. Tom doesn't play the drum. Fabienne and Tom both play with sticks.
Who plays the harp?

A. Nora ***B.*** Hana ***C.*** Fabienne
D. Leo ***E.*** Impossible to determine

4. Five ribbons have these lengths.
Which ribbon is the longest?

A. 92 dm ***B.*** 840 cm ***C.*** 88 dm ***D.*** 9400 mm ***E.*** 9 m 30 cm

5. The odometer of a car shows 567948 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

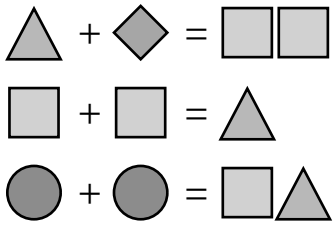
A. 1 ***B.*** 22 ***C.*** 31 ***D.*** 32 ***E.*** 34

Part B: Each question is worth 4 points

6. The seats in a theatre are numbered from 1 to 132.
How many times is the digit 4 written?

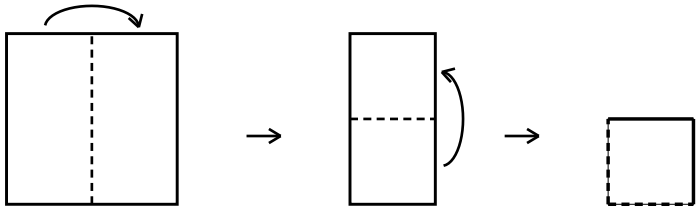
A. 27 ***B.*** 25 ***C.*** 24 ***D.*** 23 ***E.*** 13

7. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct. Which two-digit number do the cards square and diamond make?

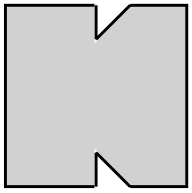
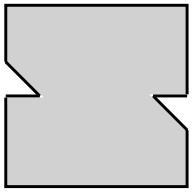
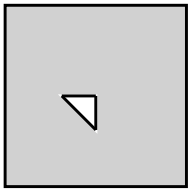
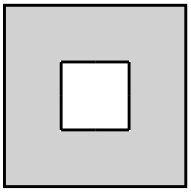
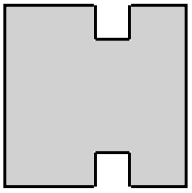


A. 10 ***B.*** 18 ***C.*** 19 ***D.*** 29 ***E.*** 91

8. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square is cut off, and the paper is unfolded.



Which design **can** be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

9. Omar was born on February 2, 2008 and Leo on August 13, 2009.
How many years old is Leo on April 15, 2015?

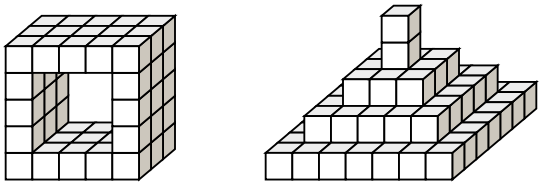
A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

10. Only the even numbers from 2 to 150 are written down: 2, 4, 6, ..., 150.
How many digits are written?

A. 75 ***B.*** 170 ***C.*** 171 ***D.*** 172 ***E.*** 174

Part C: Each question is worth 5 points

11. Mia has the tunnel on the left. Mia takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.

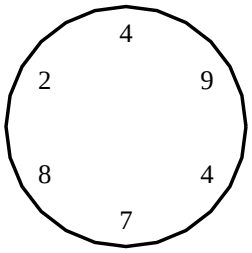


tunnel pyramid

How many more cubes does Mia need?

A. 69 ***B.*** 53 ***C.*** 39 ***D.*** 37 ***E.*** 36

12. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



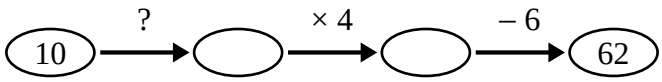
In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

13. A big cube made of small cubes is painted on the outside and taken apart. Exactly 48 small cubes have 2 painted faces.
How many small cubes was the big cube made of?

A. 343 ***B.*** 239 ***C.*** 216 ***D.*** 125 ***E.*** 64

14. The arrows show calculations done one after the other, but one arrow has lost its calculation: it shows a question mark.



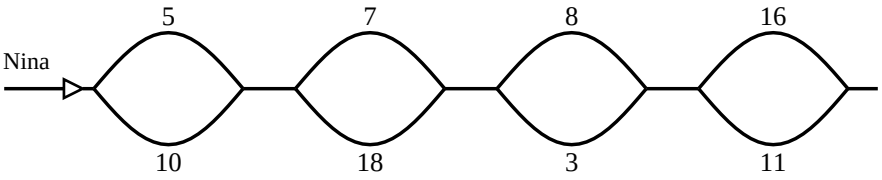
Which calculation belongs on that arrow?

A. – 4 ***B.*** + 7 ***C.*** ÷ 2 ***D.*** – 7 ***E.*** × 3

15. How many three-digit numbers have digits that add up to 7?

A. 56 ***B.*** 35 ***C.*** 33 ***D.*** 29 ***E.*** 28

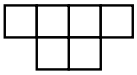
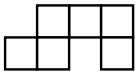
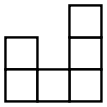
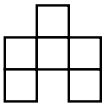
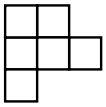
16. Nina walks along the road from left to right and adds up the numbers along the way. The total is 41.



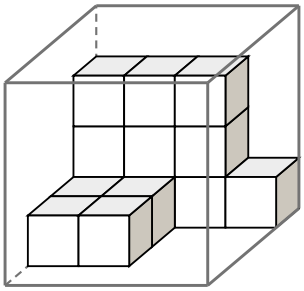
Which number did Nina see at the second fork?

A. 3 ***B.*** 5 ***C.*** 7 ***D.*** 8 ***E.*** 16

17. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

18. Leo puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Leo wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 50 ***B.*** 21 ***C.*** 20 ***D.*** 18 ***E.*** 5

19. How many different letters are used to write the two words KANGAROO and PIRATE?

A. 14 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 10

20. Five children, David, Hugo, Iris, Grace and Emma, sit in a row of 5 chairs at the cinema. David and Hugo are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 120 ***B.*** 50 ***C.*** 48 ***D.*** 24 ***E.*** 8

Answer Key

1.	**C**	**2.**	**A**	**3.**	**D**	**4.**	**D**	**5.**	**D**	**6.**	**D**
7.	**C**	**8.**	**D**	**9.**	**E**	**10.**	**D**	**11.**	**D**	**12.**	**B**
13.	**C**	**14.**	**B**	**15.**	**E**	**16.**	**C**	**17.**	**C**	**18.**	**B**
19.	**E**	**20.**	**C**								

Solutions

1. C (1+4) — The empty area has 8 squares. Pieces have 1: 4, 2: 5, 3: 4, 4: 4 squares. Only pieces 1 and 4 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · two_pieces]

2. A (8 plums) — Take 1 pineapple off both plates: 8 plums balance 2 – 1 = 1 pineapple. So one pineapple = 8 plums. [set-03/by_level/Grade_03-04/2002 Q9 · one_more_balances]

3. D (Leo) — The only possible matching: Leo – harp; Fabienne – drum; Nora – piano; Hana – flute; Tom – xylophone. [set-03/by_level/Grade_03-04/2002 Q18 · who_has]

4. D (9400 mm) — In cm: 92 dm = 920 cm, 840 cm = 840 cm, 88 dm = 880 cm, 9400 mm = 940 cm, 9 m 30 cm = 930 cm. The longest is 9400 mm. [set-03/by_level/Grade_03-04/2003 Q10 · longest]

5. D (32) — Counting up, every number before 567980 repeats a digit. 567980 has all different digits: 567980 – 567948 = 32. [set-03/by_level/Grade_03-04/2002 Q24 · all_different]

6. D (23) — As the last digit: 13 times; as the tens digit: 10 times. Total 23. [set-03/by_level/Grade_03-04/2003 Q16 · count_one_digit]

7. C (19) — Two digits add up to at most 9 + 9 = 18, so the square at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: diamond = 9, circle = 6, triangle = 2, square = 1. So the square is the tens digit and the diamond the ones digit: 19. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

8. D — A single cut appears four times, the same on the left and right and on the top and bottom. Only design D looks like that (a hole in the middle). [set-03/by_level/Grade_03-04/2007 Q24 · which_can]

9. E (5) — 2015 – 2009 = 6, but Leo's birthday (August 13) has not come yet that year, so Leo is 5. [set-03/by_level/Grade_03-04/2007 Q12 · age_on_day]

10. D (172) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 26. 4 × 1 + 45 × 2 + 26 × 3 = 172. [set-03/by_level/Grade_03-04/2007 Q23 · even_only]

11. D (37) — Tunnel: 16 × 3 = 48 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 + 1 × 1 × 2 = 85. Missing: 85 – 48 = 37. [set-03/by_level/Grade_03-04/2002 Q20 · not_enough]

12. B (1) — Each half must have 17. The ways: 4+9+4 | 7+8+2. [set-03/by_level/Grade_03-04/2002 Q11 · count_ways]

13. C (216) — A cube has 12 edges, so each edge has 48 ÷ 12 = 4 cubes between its corners. With the 2 corners, an edge is 6 cubes long, and the cube has 6 × 6 × 6 = 216. [set-03/by_level/Grade_03-04/2007 Q17 · how_big]

14. B (+ 7) — Before that arrow we have 10; working back from 62, after it we need 17. Only + 7 turns 10 into 17. [set-03/by_level/Grade_03-04/2007 Q5 · which_operation]

15. E (28) — Count by the first digit: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28. [set-03/by_level/Grade_03-04/2007 Q4 · count_with_sum]

16. C (7) — Only one route gives 41: 10 + 7 + 8 + 16 = 41. At the second fork: 7. [set-03/by_level/Grade_03-04/2007 Q1 · which_branch]

17. C — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece C fit together like that. [set-03/by_level/Grade_03-04/2007 Q8 · two_copies]

18. B (21) — Two layers hold 2 × 4 × 4 = 32 cubes. In those two layers there are already 11 cubes (columns count at most 2 each). 32 – 11 = 21. [set-03/by_level/Grade_03-04/2007 Q14 · two_layers]

19. E (10) — KANGAROO uses 6 different letters and PIRATE uses 6. A and R are in both, so count them once: 6 + 6 – 2 = 10. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]

20. C (48) — Think of David and Hugo as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block David and Hugo can swap, so 2 × 24 = 48. [set-03/by_level/Grade_03-04/2002 Q22 · side_by_side]